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Project Report

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LAUNDRYBAR SERVICE SYSTEM

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1.0 Company Background

LaundryBar was founded in 2013 by Paul Ang Beng Teck, the CEO of City Coin Laundry Sdn.Bhd, in Malaysia. The company was created to meet the growing demand for convenient, affordable and accessible laundry services, especially for the busy urban population. LaundryBar have been recognized by all the people in Malaysia today. They are also an approved franchisor by the government. The logo as shown in Figure 1, features a washing machine character holding a cloth and a blue card, paired with the bold, red-and-white text “LAUNDRYBAR” and a tagline that highlights its self-service coin laundry concept. It is an ISO Certified Company in Self Service Laundry and Maintenance.

The founder, Paul Ang Beng Teck, conducted a self-survey of the neighbourhood and identified a self-service laundry as a vital business opportunity, recognizing it as a key necessity for residents. Based on the findings from their official website, we wholeheartedly agree that LaundryBar is our first choice, particularly focusing on the branch located in Jalan Sultan Zainal Abidin, Kuala Terengganu which can be referred in Figure 2. This branch, which employs a dedicated team of 5 staff members, as shown in Figure 3, exemplifies the most promising self-service laundry concept in Malaysia. The organizational chart, as shown in Figure 4, illustrates the management team structure of City Coin LaundryBar Sdn Bhd.



Figure 1: Logo of LaundryBar

1.1 Company Objective

1. To provide twenty-four-hour self-service laundry facilities, offering customers an easy and flexible way to handle their laundry needs at any time.
2. To expand its brand through franchise opportunities, allowing entrepreneurs to own and operate their own LaundryBar outlets while maintaining consistent service quality.
3. To implement eco-friendly practices, using energy-efficient machines and environmentally safe detergents to reduce its environmental impact while providing reliable service.

1.2 Company Operations

LaundryBar Service System features a convenient machine for exchanging money into laundry coins, as well as dispensers for detergent and softener, making it easy for customers to get everything they need on-site. The branch offers a total of 9 washers and 6 dryers, all with the same capacity, ensuring ample and efficient service to accommodate customers during peak hours. Designed as an open-air facility, the branch provides a comfortable waiting area for customers. It includes seating where visitors can relax while their laundry is in progress. To further enhance convenience, there are also wheeled baskets available for customers to easily transfer their clothes from the washers to the dryers.

For optimal operation, the branch has a monthly maintenance budget of approximately RM2,000, ensuring that all machines are in top condition. A maintenance team visits the branch twice a month for thorough checkups, addressing any technical issues and keeping the facility clean and efficient.

Customers can easily contact the Kuala Terengganu branch at 1700-81-9017 or via email at laundrybaroutlet@gmail.com for inquiries or assistance. With over 350 outlets nationwide and a zero-failure rate, LaundryBar's success in this specific location highlights its potential. Furthermore, the company has expanded into countries like Thailand, Vietnam, and Indonesia, underscoring its strong growth trajectory.





Figure 2: Branch in Jalan Sultan Zainal Abidin

1.3 Company Structure



Figure 3: LaundryBar Organizational Structure

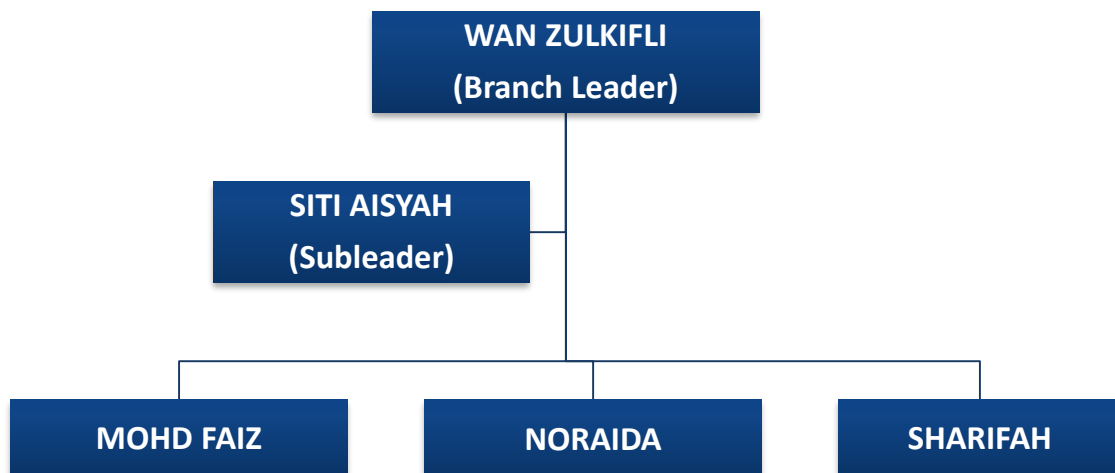


Figure 4: Branch's Organizational Structure

2.0 Current System Description

LaundryBar's current system relies on manual and semi-automated processes. When a customer enters the laundromat, they select a machine based on availability and proceed to the token machine by inserting cash which converts it into tokens. One token is equal to RM1. The calculation process for both machines, washer and dryer depends on the selected temperature which are low, medium, or high. The default cycle duration is 26 minutes, and additional time can be purchased in increments of 6 minutes up to a maximum of 18 minutes. For instance, it will be limited to 6 min, 12 min, or 18 min. The machine is activated after the required number of tokens is inserted. Customers can either bring and use their own laundry products, such as detergent and softener, or they can opt to use the laundry care products provided by the laundromat. The system logs the usage of the laundromat-provided products. Customers may wait on-site while the laundry cycle runs or leave and return later. Once the cycle is complete, they collect their clothes.

If any machine requires maintenance, the issue is logged manually. The shop updates maintenance schedules and inventory levels at the end of the day, usually through semi-automated system which may be done manually. Customers use POS systems for payments, but transaction records, machine maintenance tracking, and supply management are often maintained through manual methods like spreadsheets. There's no centralized system to manage data from multiple outlets, which can cause delays in updating maintenance schedules or inventory levels.

2.1 Flowchart of the Current System Description

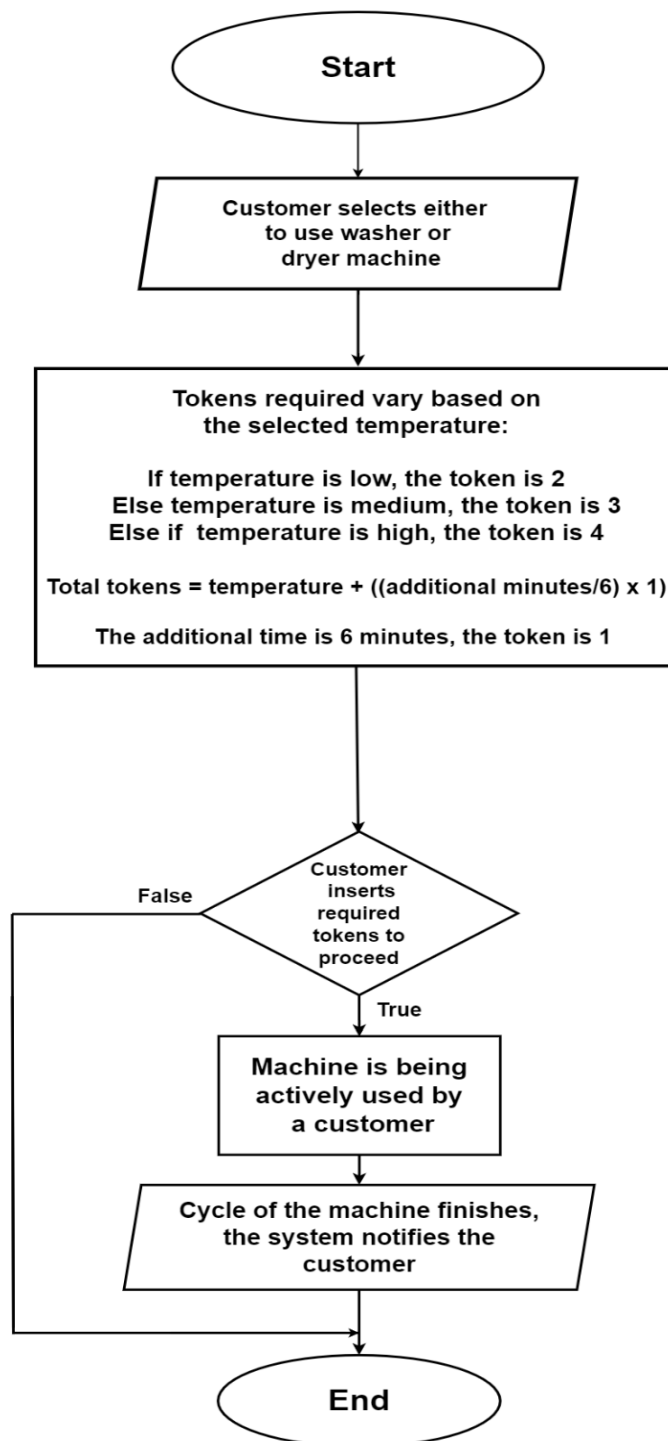


Figure 5: The System's Flowchart

3.0 Problem Statements

Relying on a traditional coin-based payment system in LaundryBar presents a variety of issues that impact customer experience and operational efficiency. For customers, the need to use coins adds inconvenience and can cause payment delays and errors, especially when change is not available or when machines reject certain coins. In an era where most consumers choose modern and flexible payment options such as mobile payments and credit cards, being limited to coins restricts customer options, creating an uncomfortable and ultimately unsatisfactory experience that can undermine customer loyalty. Operationally, coin-based systems complicate monthly sales tracking, as staff must manually collect and count coins from each machine as well as a labor-intensive process prone to human error. These errors can lead to discrepancies, complicate accurate financial tracking and slow down the reporting process, which LaundryBar owners rely on for timely business insights. Without automated or digital alternatives, these limitations reduce the ability of businesses to monitor performance in real time, make rapid operational decisions and plan effectively. Therefore, reliance on this outdated system not only reduces customer satisfaction but also constrains LaundryBar's efficiency and potential to grow.

Other than that, LaundryBar's lack of an efficient system to track machine performance leads to frequent breakdowns and maintenance problems, which significantly affects service quality and customer satisfaction. Without a reliable method to monitor the condition of each machine, problems often go unnoticed until they become severe, resulting in unexpected breakdowns and reduced machine life. This reactive approach not only disrupts LaundryBar operations but also inconveniences customers who expect a functional and reliable machine to complete their laundry efficiently and quickly. When machines underperform or break down, customers face additional frustrations, such as incomplete washes, delays or damage to their clothes, leading to a reduced overall experience. Additionally, without insight into machine usage patterns, LaundryBar misses opportunities for proactive maintenance, which can prevent wear and tear and

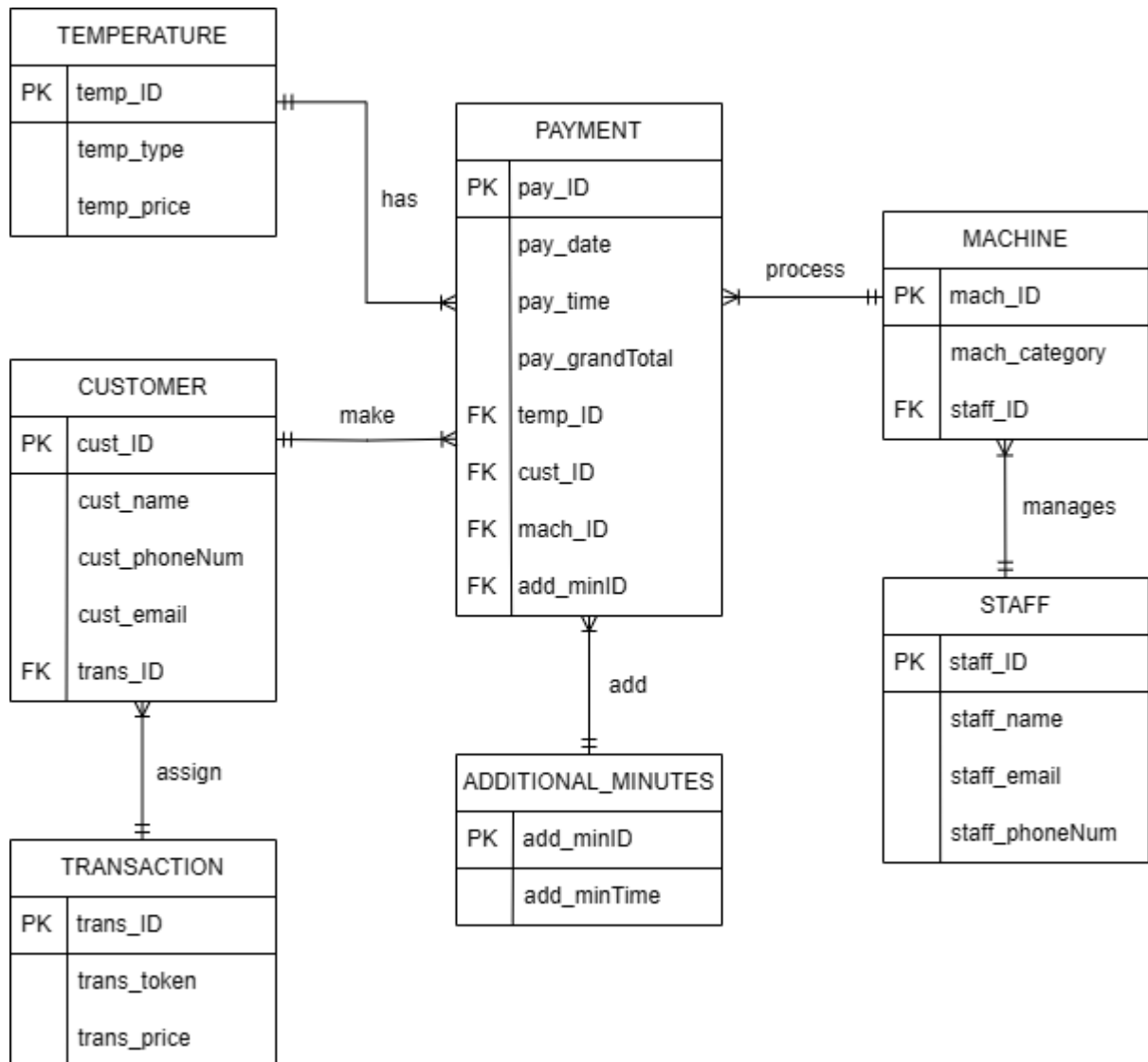
extend machine life. The absence of such a system also hinders the ability to maintain a regular and cost-effective maintenance schedule, leading to increased operating costs over time. Therefore, implementing a performance tracking system to monitor machine health and schedule preventive maintenance will not only improve operational efficiency but also ensure a high-quality experience for customers, supporting customer loyalty and business profitability.

Additionally, LaundryBar seeks to enhance its operational efficiency and improve customer satisfaction data-driven decision-making. Currently, LaundryBar only offers three standard water temperature options which are hot, medium and cold causing customers to often face significant limitations in meeting their specific laundry needs. While these basic settings meet general needs, such as using hot water for heavily soiled items and cold water for delicate fabrics, many laundry loads require more precise temperature control for optimal results. For example, certain stains, such as grease or protein-based marks, may require a specific temperature range to be removed effectively, and relying solely on standard options may result in insufficient cleaning. Additionally, some types of fabric, such as wool or silk, require the right temperature to prevent damage, which may not be accommodated by available options. Furthermore, as customers increasingly prioritize energy efficiency and sustainability, a lack of flexibility in water temperature settings can lead to higher energy consumption, as customers may use hotter washes to achieve the desired cleaning results. This limitation not only reduces overall customer satisfaction but also increases operating costs for LaundryBar owners, who may struggle to attract and retain customers looking for a more customized and efficient laundry experience. Finally, by failing to provide a wider range of temperature options, LaundryBar could risk negatively impacting their business results.

4.0 Objectives

1. To transform manual file system into a database for LaundryBar Service System.
2. To create a database system for LaundryBar Service System.
3. To generate reports for LaundryBar Service System.

5.0 Entity Relationship Diagram (ERD)



6.0 3NF Relational Schema

CUSTOMER (cust_ID, cust_name, cust_phoneNum, cust_email, trans_ID*)

TRANSACTION (trans_ID, trans_token, trans_price)

MACHINE (mach_ID, mach_category, staff_ID*)

STAFF (staff_ID, staff_name, staff_email, staff_phoneNum)

PAYMENT (pay_ID, pay_date, pay_time, pay_grandTotal, temp_ID*, cust_ID*,
mach_ID*, add_minID*)

TEMPERATURE (temp_ID, temp_type, temp_price)

ADDITIONAL_MINUTES (add_minID, add_minTime)

7.0 Data Dictionary

Table Name	Attribute	Description	Data Type	Sample Data	PK or FK
CUSTOMER	cust_ID	Unique identifier foreach customer	Short text	99	PK
	cust_name	Full name of thecustomer	Long text	Lee Zii Jia	
	cust_phoneNum	Customer's phone number	Short text	011-9287302	
	cust_email	Customer's email address	Long text	jia@gmail.com	
	trans_ID	Unique identifier for each transaction	Short text	TR01	FK

Table Name	Attribute	Description	Data Type	Sample Data	PK or FK
TRANSACTION	trans_ID	Unique identifier foreach transaction	Short text	TR01	PK
	trans_token	Number of tokens or items used in the transaction	Number	1	
	trans_price	The total cost incurred in the transaction	Currency	RM1.00	

Table Name	Attribute	Description	Data Type	Sample Data	PK or FK
MACHINE	mach_ID	Unique identifier foreach machine	Short text	D1	PK
	mach_category	Category or type of the machine (e.g., dryer, washer)	Short text	Dryer	
	staff_ID	Unique identifier for each staff members	Short text	S01	FK

Table Name	Attribute	Description	Data Type	Sample Data	PK or FK
STAFF	staff_ID	Unique identifier foreach staff members	Short text	S01	PK
	staff_name	Full name of the staff members	Short text	Wan Zukkifli	
	staff_email	Staff's email address	Short text	Wz@gmail.com	
	staff_phoneNum	Staff's phone number	Short text	012-6537786	

Table Name	Attribute	Description	Data Type	Sample Data	PK or FK
PAYMENT	pay_ID	Unique identifier for each payment	Short text	P01	PK
	pay_date	Date when thepayment was made	Date/Time	12/8/2024	
	pay_time	Time when thepayment was completed	Date/Time	2:52:02 AM	
	pay_grandTotal	Grand total of the service fees	Currency	RM5.00	
	temp_ID	Unique identifier for each temperature	Short text	TH	FK
	cust_ID	Unique identifier foreach customer	Short text	99	FK
	mach_ID	Unique identifier for each machine	Short text	W1	FK
	add_minID	Unique identifier for each additional minute	Short text	A1	FK

Table Name	Attribute	Description	Data Type	Sample Data	PK or FK
TEMPERATURE	temp_ID	Unique identifier foreach temperature	Short text	TH	PK
	temp_type	Type of temperature of the machine (e.g., low, medium, high)	Short text	High	
	temp_price	The total cost incurred for each temperature	Number	4	

Table Name	Attribute	Description	Data Type	Sample Data	PK or FK
ADDITIONAL_MINUTES	add_minID	Unique identifier foreach additional minute	Short text	A1	PK
	add_minTime	The choices and their prices to add minutes	Number	6	

8.0 Sample Output

8.1 Tables

CUSTOMER

CUSTOMER X					
	cust_ID ▾	cust_name ▾	cust_phoneNum ▾	cust_email ▾	trans_ID ▾
+	01	Ahmad Ali	010-1234567	ahmad.ali@gmail.com	TR01
+	02	Siti Nur	011-2345678	siti_nur2000@gmail.com	TR01
+	03	Mohd Zainal	012-3456789	zainal_mhd@yahoo.com	TR01
+	04	Farah Aziz	013-4567890	farah.aziz77@hotmail.com	TR01
+	05	Hassan Ibrahim	011-8765432	hassan.ibrahim22@yahoo.com	TR01
+	06	Noraida Rahman	010-9876543	noraida_rahman@gmail.com	TR01
+	07	Zulfiqar Jamal	012-2345678	zulfiqar.jamal12@gmail.com	TR01
+	08	Aminah Amin	013-3456789	aminah_amin21@yahoo.com	TR01
+	09	Rashidah Mohamed	010-6543210	rashidah.mohd24@gmail.com	TR01
+	10	Syarif Ali	011-7654321	syarif.ali_89@outlook.com	TR01
+	11	Shah Ibrahim	012-5678901	shah.m@icloud.com	TR01
+	12	Ayu Hassan	013-6789012	ayu_hassan03@gmail.com	TR01
+	13	Fazilah Zulkifli	010-1239876	fazilah.zulkifli_89@outlook.com	TR01
+	14	Rizaluddin Omar	011-2346789	rizal.omar_98@yahoo.com	TR01
+	15	Khalid Arif	012-3451234	khalidarif_75@gmail.com	TR01
+	16	Roslina Jamil	013-4562345	roslina.jamil99@gmail.com	TR01
+	17	Fahmi Yusuf	010-6544321	fahmi_yusuf123@gmail.com	TR01
+	18	Lina Rani	011-7659876	lina.rani@hotmail.com	TR01
+	19	Lee Wei	012-3456789	lee.wei_85@gmail.com	TR01
+	20	Tan Mei Ling	011-9876543	tan.meiling@yahoo.com	TR01
+	21	Wong Jun Jie	016-1029384	wong.junjie_76@hotmail.com	TR01

TRANSACTION

TRANSACTION X					
	trans_ID ▾	trans_token ▾	trans_price ▾	Click to Add ▾	
+	TR01	1	RM1.00		
*		0	RM0.00		

MACHINE

MACHINE X						
	mach_ID	mach_category	staff_ID	Click to Add		
+	D1	Dryer	S01			
+	D2	Dryer	S01			
+	D3	Dryer	S01			
+	D4	Dryer	S01			
+	D5	Dryer	S01			
+	W1	Washer	S02			
+	W2	Washer	S02			
+	W3	Washer	S02			
+	W4	Washer	S02			
+	W5	Washer	S02			

STAFF

STAFF X						
	staff_ID	staff_name	staff_email	staff_phoneNum	Click to Add	
+	S01	Wan Zulkifli	wan1@gmail.com	012-6537786		
+	S02	Siti Aisyah	aisyah1988@gmail.com	011-5903442		

PAYMENT

PAYMENT X								
	pay_ID	pay_date	pay_time	pay_grandTotal	temp_ID	cust_ID	mach_ID	add_minID
	P01	12/8/2024	3:52:02 AM	RM5.00	TH	01	D1	A1
	P02	1/12/2024	12:00:04 PM	RM5.00	TH	02	W1	A1
	P03	23/3/2024	4:58:27 PM	RM3.00	TM	03	W1	
	P04	15/9/2024	9:12:04 PM	RM4.00	TL	04	W2	A2
	P05	21/7/2024	6:58:29 PM	RM4.00	TH	05	W5	
	P06	21/7/2024	8:02:55 PM	RM3.00	TL	05	D3	A1
	P07	12/8/2024	12:24:55 AM	RM5.00	TM	06	W5	A2
	P08	12/8/2024	12:27:00 AM	RM3.00	TM	06	W3	
	P09	12/8/2024	1:15:17 AM	RM3.00	TM	06	D5	
	P10	25/3/2024	4:50:03 PM	RM3.00	TL	07	W3	A1
	P100	17/2/2024	1:17:08 AM	RM4.00	TH	74	W4	
	P101	18/2/2024	11:09:12 AM	RM2.00	TL	75	W4	
	P102	17/5/2024	2:45:11 PM	RM3.00	TM	76	W1	
	P103	19/4/2024	3:55:14 PM	RM7.00	TH	77	W2	A3
	P104	10/7/2024	10:58:17 AM	RM2.00	TL	78	W2	
	P105	10/7/2024	11:01:00 AM	RM3.00	TM	78	W3	
	P106	10/7/2024	12:01:11 PM	RM4.00	TH	78	D4	
	P107	10/1/2024	12:34:45 PM	RM3.00	TM	79	W5	
	P108	10/11/2024	12:03:12 AM	RM2.00	TL	80	D1	
	P109	10/12/2024	3:45:23 PM	RM6.00	TH	81	W4	A2
	P11	25/3/2024	5:17:11 PM	RM2.00	TL	07	D2	

TEMPERATURE

TEMPERATURE X							
	temp_ID ▾	temp_type ▾	temp_price ▾	Click to Add ▾			
	TH	High	4				
	TL	Low	2				
	TM	Medium	3				

ADDITIONAL_MINUTES

ADDITIONAL_MINUTES X							
	add_minID ▾	add_minTime ▾	Click to Add ▾				
	A1	6					
	A2	12					
	A3	18					

8.2 Entity Relationship Diagram (ERD)

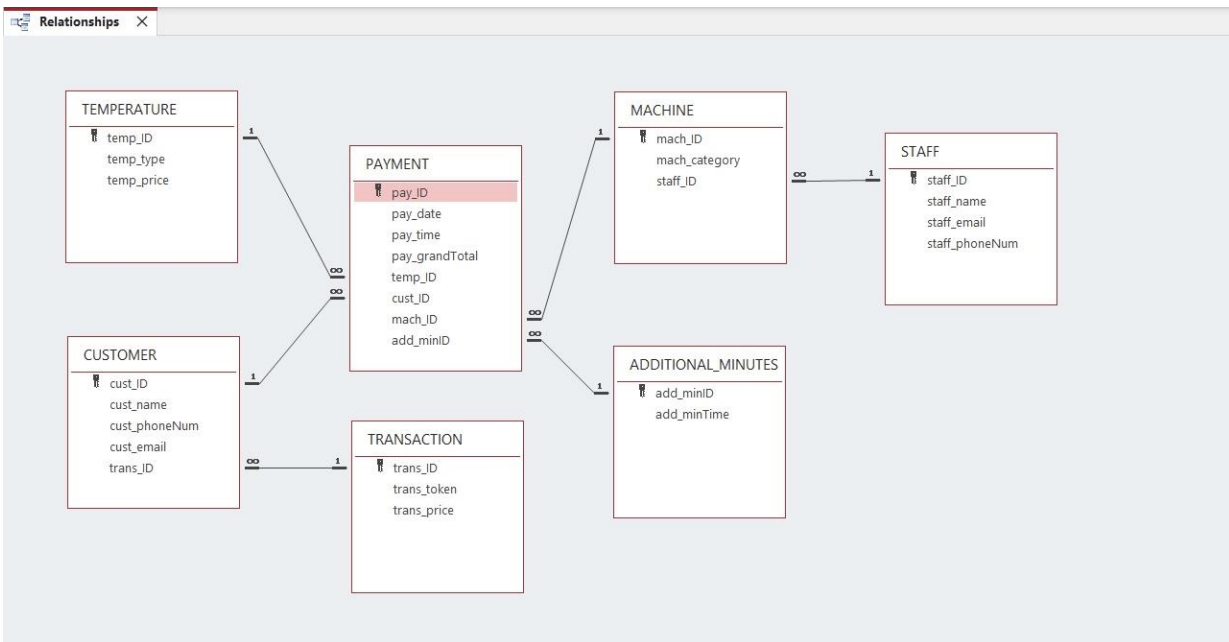


Figure 8.2.1: Entity Relationship Diagram (ERD) of LaundryBar Service System

8.3 Forms

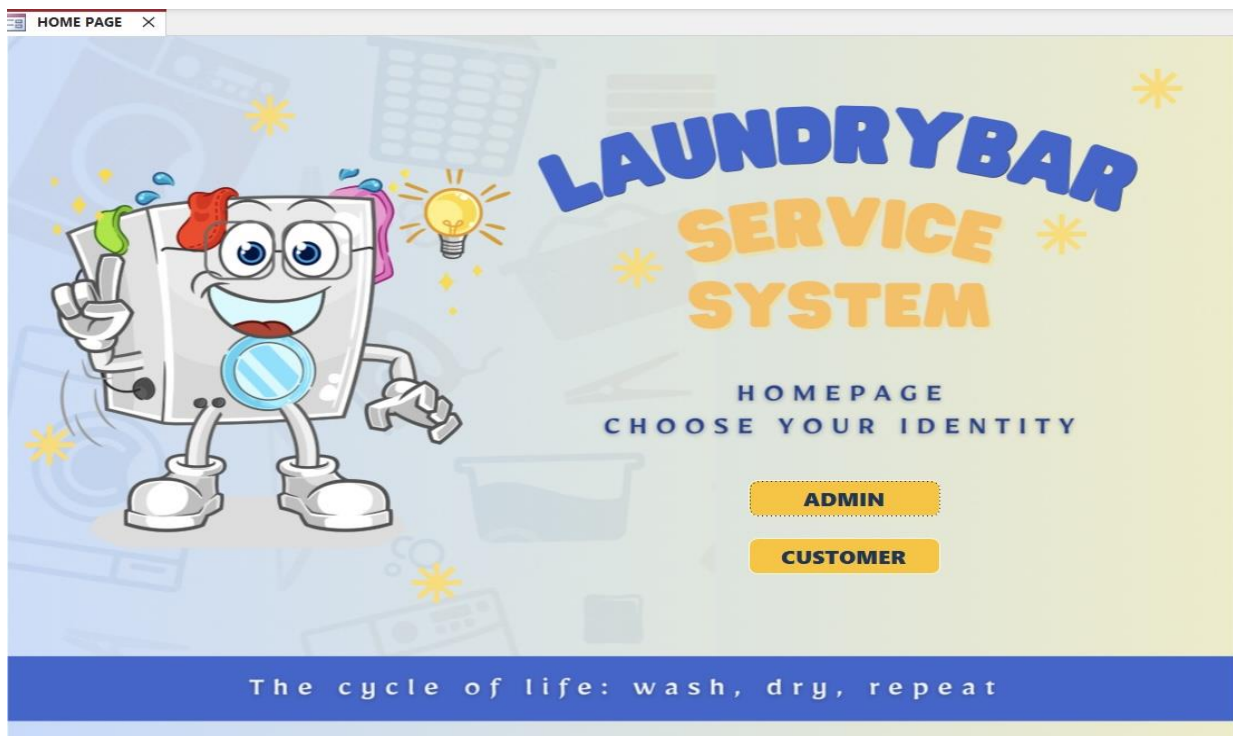


Figure 8.3.1: Home page for this system

This form is for user to choose either they are customer or admin

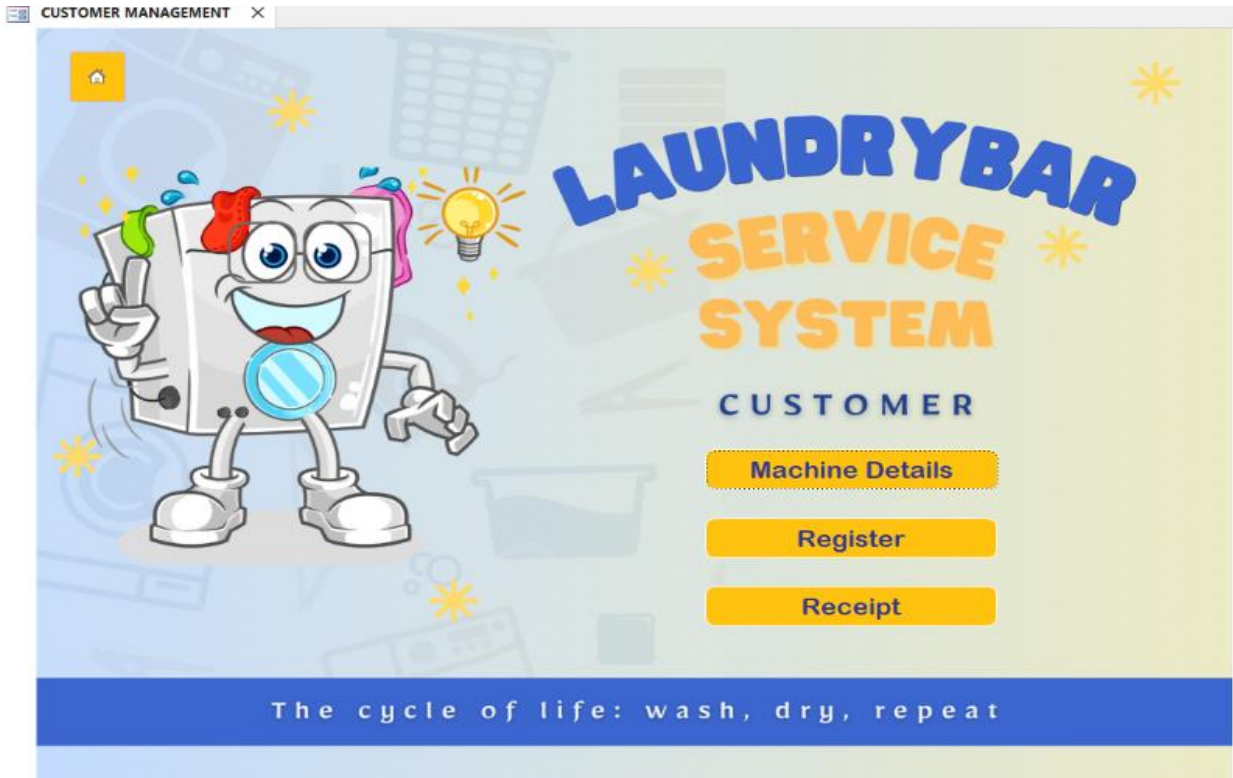


Figure 8.3.2: Home page for customer

This form is for customer to choose the machine details or receipt

MACHINE

MACHINE ID: New Machine

CATEGORY:

STAFF IN CHARGE:

TEMPERATURE DETAILS

temp_ID	temp_type	temp_price
TH	High	4
TL	Low	2
TM	Medium	3
		0

Record: 14 of 3 No Filter Search

ADDITIONAL MINUTES DETAILS

add_minID	add_minTime
A1	6
A2	12
A3	18
	0

Record: 14 of 3 No Filter Search

Figure 8.3.3: Machine form

This form is to be used for customer to search machine

CUSTOMER1 X

 **NEW CUSTOMER**  

CUSTOMER ID

NAME

PHONE NO.

EMAIL

TRANS ID Note: trans ID = TR01

Figure 8.3.4: New Customer form

This form is to let customer make new registration

receipt X

 **RECEIPT** 

CUSTOMER ID

PAYMENT ID

TIME

DATE

TOTAL PAYMENT

STAFF IN CHARGE

Figure 8.3.5: Receipt form

This form is to be used for customer to view their receipt

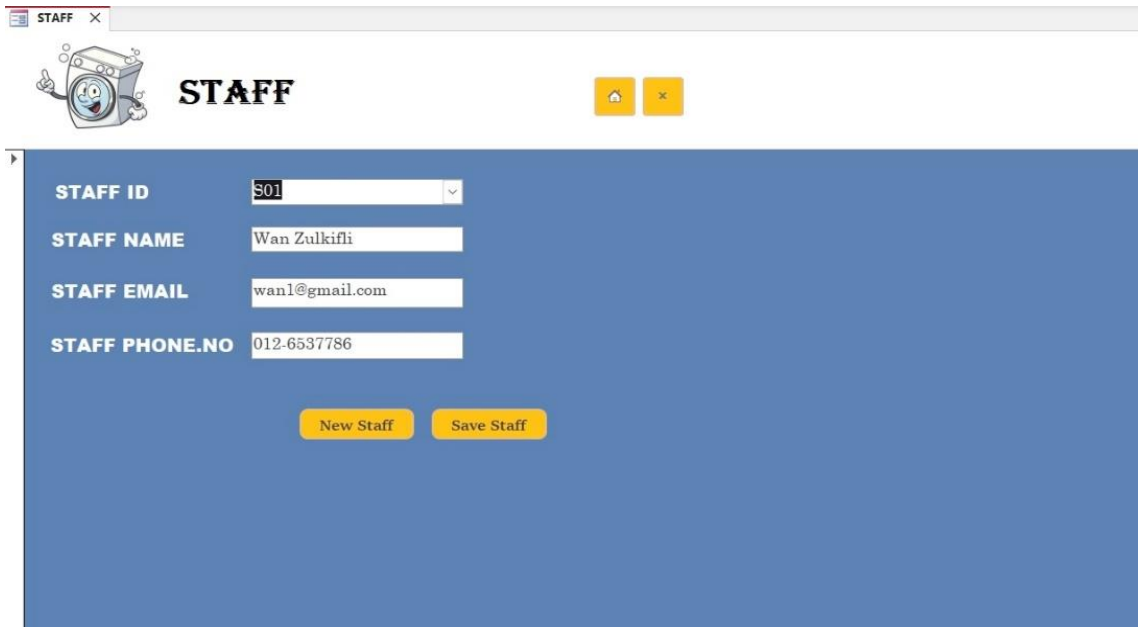


Figure 8.3.6: Home page for admin

This form is to be used for admin to choose either Customer Record, Customer List, Staff Record or Machine List

Figure 8.3.7: Customer form for admin

This form is to be used for admin to know the details of customer selected



STAFF

STAFF ID: S01

STAFF NAME: Wan Zulkifli

STAFF EMAIL: wan1@gmail.com

STAFF PHONE.NO: 012-6537786

New Staff Save Staff

Figure 8.3.8: Staff form for admin

This form is to be used for admin to know the details of staff selected

8.4 Report

 CUSTOMER ORDER HISTORY								
CUSTOMER ID	NAME	PHONE NO.	PAYMENT I	DATE	TIME	GRAND TOTAL	MACHINE ID	CATEGORY
01	Ahmad Ali	011-0945334	P01	12/8/2024	3:52:02 AM	RM5.00	D1	Dryer
02	Siti Nur	011-2345678	P02	1/12/2024	12:00:04 PM	RM5.00	W1	Washer
03	Mohd Zainal	012-3456789	P03	23/3/2024	4:58:27 PM	RM3.00	W1	Washer
04	Farah Aziz	013-4567890	P04	15/9/2024	9:12:04 PM	RM4.00	W2	Washer
05	Hassan Ibrahim	011-8765432	P05	21/7/2024	6:58:29 PM	RM4.00	W5	Washer
Sunday, 19 January, 2025								
Page 1 of 17								

Figure 8.4.1: Report for customer order history

This report is to show the history of customer's machine usage

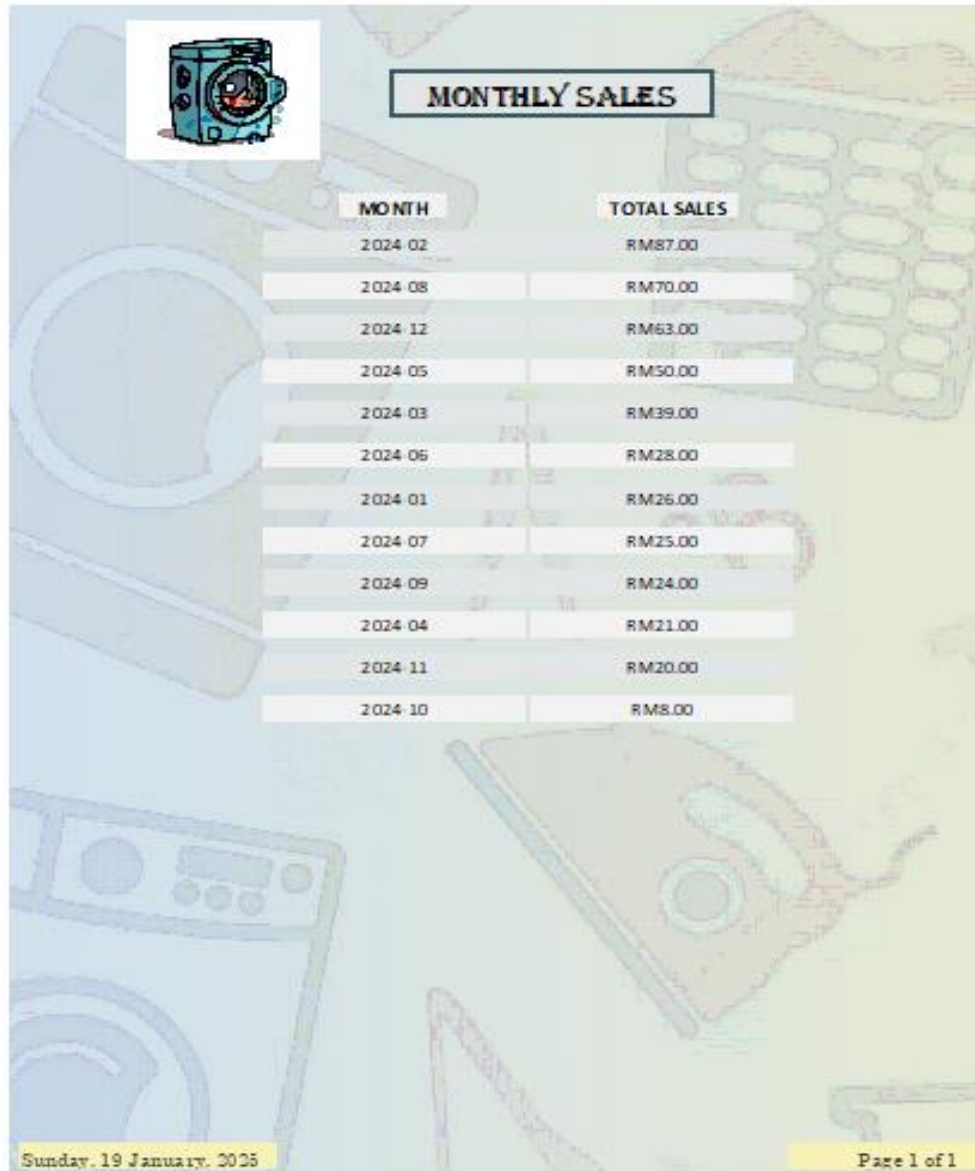


Figure 8.4.2: Report for monthly sales

This report is to show the highest to lowest total sales of LaundryBar for each month

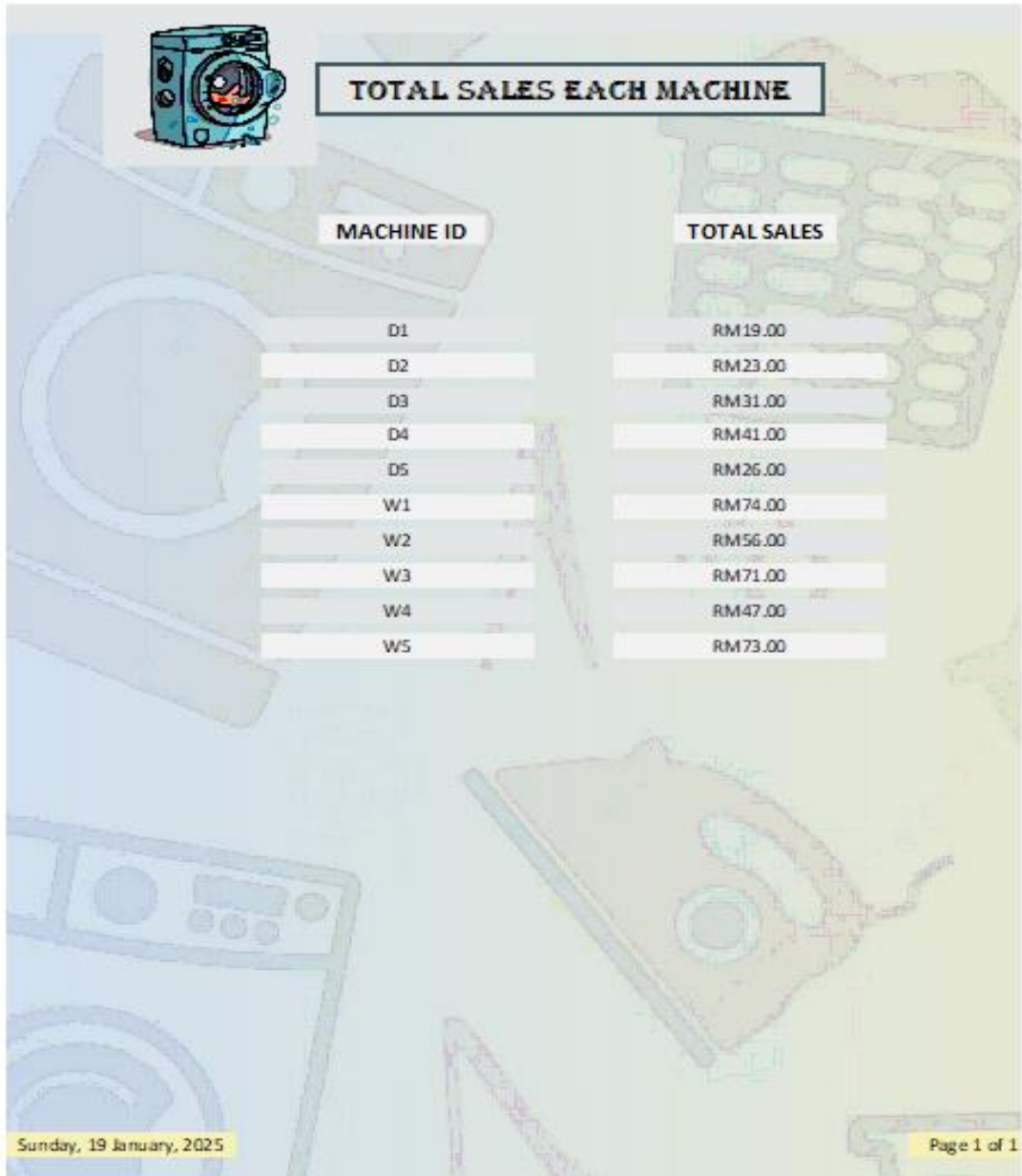


Figure 8.4.3: Report for total sales for each machine
This report is to show the total sales for each machine

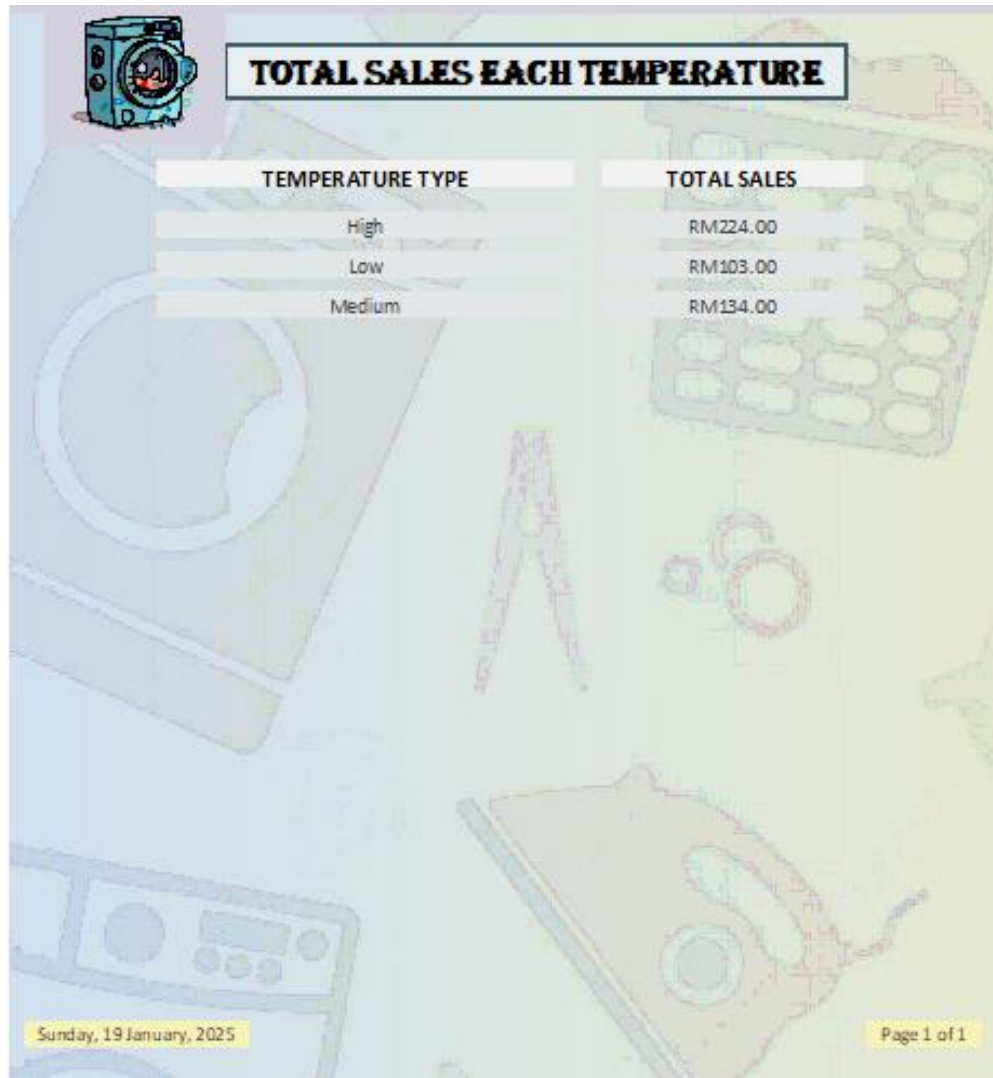


Figure 8.4.4: Report for total sales for each temperature
This report is to show the total sales for each temperature



RECEIPT

CUSTOMER ID	99
PAYMENT ID	P127
PAYMENT TIME	6:55:33 AM
PAYMENT DATE	7/2/2024
PAYMENT GRANDTOTAL	RM5.00
STAFF ID	S02

Sunday, 19 January, 2025

Page 1 of 1

Figure 8.4.5: Report for receipt

This report is to show the receipt for a customer based on their ID

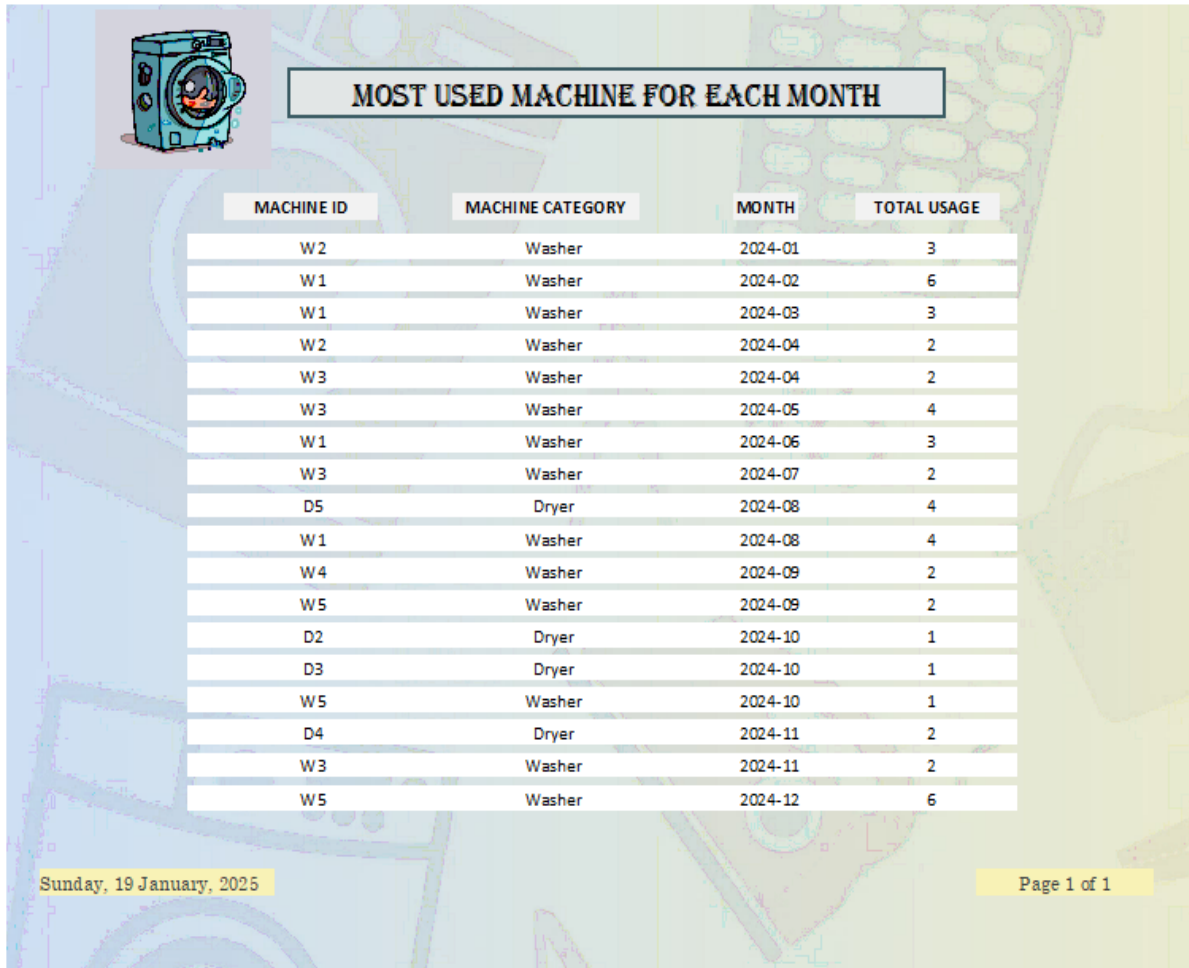


Figure 8.4.6: Report for most used machine

This report is to show the report of most used machine for each month

9.0 SQL

9.0.1 The total average payment made by customers

Average payment amount X					
SELECT AVG(pay_grandTotal) AS average_payment FROM PAYMENT;					
Average payment amount X					
average_payment					
RM3.55					

9.0.2 The total average payment for each machine

Average payment each machine X					
SELECT Machine.mach_ID, AVG(Payment.pay_grandTotal) AS average_payment FROM Machine, Payment WHERE Machine.mach_ID = Payment.mach_ID AND Machine.mach_ID IN ('W1', 'W2', 'W3', 'W4', 'W5', 'D1', 'D2', 'D3', 'D4', 'D5') GROUP BY Machine.mach_ID;					
Average payment each machine X					
mach_ID		average_payment			
D1		RM3.17			
D2		RM2.88			
D3		RM3.44			
D4		RM4.10			
D5		RM3.71			
W1		RM3.70			
W2		RM3.73			
W3		RM3.23			
W4		RM3.92			
W5		RM3.48			

9.0.3 The history of customer's order

Customer order history X									
SELECT c.cust_ID, c.cust_name, c.cust_phoneNum, p.pay_ID, p.pay_date, p.pay_time, p.pay_grandTotal, m.mach_ID, m.mach_category FROM CUSTOMER AS c, PAYMENT AS p, MACHINE AS m WHERE c.cust_ID = p.cust_ID AND p.mach_ID = m.mach_ID ORDER BY c.cust_ID, p.pay_date, p.pay_time;									
Customer order history X									
cust_ID	cust_name	cust_phoneNum	pay_ID	pay_date	pay_time	pay_grandTotal	mach_ID	mach_category	
01	Ahmad Ali	010-1234567	P01	12/8/2024	3:52:02 AM	RM5.00	D1	Dryer	
02	Siti Nur	011-2345678	P02	1/12/2024	12:00:04 PM	RM5.00	W1	Washer	
03	Mohd Zainal	012-3456789	P03	23/3/2024	4:58:27 PM	RM3.00	W1	Washer	
04	Farah Aziz	013-4567890	P04	15/9/2024	9:12:04 PM	RM4.00	W2	Washer	
05	Hassan Ibrahim	011-8765432	P05	21/7/2024	6:58:29 PM	RM4.00	W5	Washer	
05	Hassan Ibrahim	011-8765432	P06	21/7/2024	8:02:55 PM	RM3.00	D3	Dryer	
06	Noraida Rahman	010-9876543	P07	12/8/2024	12:24:55 AM	RM5.00	W5	Washer	
06	Noraida Rahman	010-9876543	P08	12/8/2024	12:27:00 AM	RM3.00	W3	Washer	
06	Noraida Rahman	010-9876543	P09	12/8/2024	1:15:17 AM	RM3.00	D5	Dryer	
07	Zulfiqar Jamal	012-2345678	P10	25/3/2024	4:50:03 PM	RM3.00	W3	Washer	
07	Zulfiqar Jamal	012-2345678	P11	25/3/2024	5:17:11 PM	RM2.00	D2	Dryer	
08	Aminah Amin	013-3456789	P12	25/1/2024	5:17:11 PM	RM2.00	W2	Washer	
09	Rashidah Mohamed	010-6543210	P13	19/12/2024	7:23:11 PM	RM2.00	W3	Washer	
10	Syarif Ali	011-7654321	P130	25/3/2024	12:18:22 PM	RM4.00	D4	Dryer	
10	Syarif Ali	011-7654321	P129	25/3/2024	11:25:01 PM	RM2.00	W3	Washer	
10	Syarif Ali	011-7654321	P14	15/6/2024	1:16:55 PM	RM4.00	W2	Washer	
10	Syarif Ali	011-7654321	P15	15/6/2024	2:34:23 PM	RM2.00	D1	Dryer	
11	Shah Ibrahim	012-5678901	P16	23/4/2024	9:43:12 PM	RM3.00	W3	Washer	
12	Ayu Hassan	013-6789012	P17	29/6/2024	3:21:45 PM	RM2.00	W3	Washer	
13	Fazilah Zulkifli	010-1239876	P18	12/8/2024	12:54:21 PM	RM5.00	W4	Washer	
13	Fazilah Zulkifli	010-1239876	P19	12/8/2024	1:55:43 PM	RM3.00	D5	Dryer	

9.0.4 The machine used within period entered by customers

Machine used within period X				
<pre> 1 SELECT MACHINE.mach_ID, MACHINE.mach_category, PAYMENT.pay_date, COUNT(MACHINE.mach_category) AS Total 2 FROM PAYMENT, MACHINE 3 WHERE PAYMENT.mach_ID = MACHINE.mach_ID 4 AND PAYMENT.pay_date BETWEEN [START DATE] AND [END DATE] 5 GROUP BY MACHINE.mach_ID, MACHINE.mach_category, PAYMENT.pay_date 6 ORDER BY PAYMENT.pay_date; 7 </pre>				
Machine used within period X				
mach_ID	mach_category	pay_date	Total	
D5	Dryer	1/1/2024	1	
W2	Washer	5/1/2024	1	
W5	Washer	10/1/2024	1	
D2	Dryer	16/1/2024	2	
W2	Washer	16/1/2024	1	
W4	Washer	16/1/2024	1	
W2	Washer	25/1/2024	1	
W5	Washer	1/2/2024	1	
W1	Washer	2/2/2024	1	

9.0.5 The maximum payment made by customers

Max payment by customer X				
<pre> SELECT Customer.cust_ID, Payment.pay_grandTotal FROM Customer, Payment WHERE Customer.cust_ID = Payment.cust_ID AND Payment.pay_grandTotal = (SELECT MAX(pay_grandTotal) FROM Payment); </pre>				
Max payment by customer X				
cust_ID	pay_grandTotal			
52	RM7.00			
71	RM7.00			
77	RM7.00			
87	RM7.00			
92	RM7.00			

9.0.6 The monthly sales in descending order

	Monthly sales highest to lowest	✕
<pre>SELECT FORMAT(pay_date, "yyyy-mm") AS payment_month, SUM(pay_grandTotal) AS total_payment FROM Payment GROUP BY FORMAT(pay_date, "yyyy-mm") ORDER BY SUM(pay_grandTotal) DESC;</pre>		

	Monthly sales highest to lowest	✕
payment_month	total_payment	
2024-02	RM87.00	
2024-08	RM70.00	
2024-12	RM63.00	
2024-05	RM50.00	
2024-03	RM39.00	
2024-06	RM28.00	
2024-01	RM26.00	
2024-07	RM25.00	
2024-09	RM24.00	
2024-04	RM21.00	
2024-11	RM20.00	
2024-10	RM8.00	

9.0.7 The most used machine for each month

Most used machine each month X	
<pre> SELECT Format(PAYMENT.pay_date, "yyyy-mm") AS year_month, MACHINE.mach_ID, MACHINE.mach_category, COUNT(PAYMENT.pay_ID) AS total_usage FROM PAYMENT, MACHINE WHERE PAYMENT.mach_ID = MACHINE.mach_ID GROUP BY Format(PAYMENT.pay_date, "yyyy-mm"), MACHINE.mach_ID, MACHINE.mach_category HAVING COUNT(PAYMENT.pay_ID) = (SELECT MAX(UsageCount) FROM (SELECT Format(P.pay_date, "yyyy-mm") AS SubMonthYear, P.mach_ID, COUNT(P.pay_ID) AS UsageCount FROM PAYMENT AS P GROUP BY Format(P.pay_date, "yyyy-mm"), P.mach_ID) AS SubQuery WHERE SubQuery.SubMonthYear = Format(PAYMENT.pay_date, "yyyy-mm")) ORDER BY Format(PAYMENT.pay_date, "yyyy-mm"); </pre>	

Most used machine each month					
year_month	mach_ID	mach_category	total_usage		
2024-01	W2	Washer	3		
2024-02	W1	Washer	6		
2024-03	W1	Washer	3		
2024-04	W2	Washer	2		
2024-04	W3	Washer	2		
2024-05	W3	Washer	4		
2024-06	W1	Washer	3		
2024-07	W3	Washer	2		
2024-08	D5	Dryer	4		
2024-08	W1	Washer	4		
2024-09	W4	Washer	2		
2024-09	W5	Washer	2		
2024-10	D2	Dryer	1		
2024-10	D3	Dryer	1		
2024-10	W5	Washer	1		
2024-11	D4	Dryer	2		
2024-11	W3	Washer	2		
2024-12	W5	Washer	6		

9.0.8 The most used temperature chosen by customers

Most used temperature X					
SELECT Temperature.temp_type, COUNT(Payment.pay_ID) AS usage_count FROM Temperature, Payment WHERE Temperature.temp_ID = Payment.temp_ID GROUP BY Temperature.temp_type ORDER BY COUNT(Payment.pay_ID) DESC;					
temp_type	usage_count				
High	48				
Low	43				
Medium	39				

9.0.9 Receipt for each customer based on selected ID

Receipt X					
SELECT CUSTOMER.cust_ID, PAYMENT.pay_ID, PAYMENT.pay_time, PAYMENT.pay_date, PAYMENT.pay_grandTotal, MACHINE.staff_ID FROM CUSTOMER, PAYMENT, MACHINE, STAFF WHERE CUSTOMER.cust_ID = PAYMENT.cust_ID AND PAYMENT.mach_ID = MACHINE.mach_ID AND MACHINE.staff_ID = STAFF.staff_ID AND PAYMENT.cust_ID = [Input Customer ID];					
cust_ID	pay_ID	pay_time	pay_date	pay_grandTotal	staff_ID
01	P01	3:52:02 AM	12/8/2024	RM5.00	S01

9.0.10 The total additional minutes that purchased by customers

Total addmin purchased by cust	×
SELECT Payment.cust_ID, SUM(Additional_Minutes.add_minTime) AS total_additional_minutes FROM Payment, Additional_Minutes WHERE Payment.add_minID = Additional_Minutes.add_minID GROUP BY Payment.cust_ID;	

Total addmin purchased by cust					
cust_ID	total_additional_minutes				
01	6				
02	6				
04	12				
05	6				
06	12				
07	6				
10	6				
13	6				
15	6				
19	12				
23	18				
28	6				
30	12				
36	6				
38	24				
39	18				
41	12				
43	6				
46	6				
50	18				
52	18				

9.0.11 The total customers that visit LaundryBar for more than once

Total cust visits >1 X						
SELECT c.cust_ID, c.cust_name, v.TotalVisits FROM CUSTOMER AS c, (SELECT p.cust_ID, COUNT(p.pay_ID) AS TotalVisits FROM PAYMENT AS p GROUP BY p.cust_ID HAVING COUNT(p.pay_ID) > 1) AS v WHERE c.cust_ID = v.cust_ID ORDER BY v.TotalVisits DESC;						
Total cust visits >1 X						
cust_ID	cust_name	TotalVisits				
30	Lim Xiao Hui	4				
10	Syarif Ali	4				
78	Vishal Raj	3				
06	Noraida Rahman	3				
62	Vijay Desai	3				
07	Zulfiqar Jamal	2				
13	Fazilah Zulkifli	2				
14	Rizaluddin Omar	2				
15	Khalid Arif	2				
18	Lina Rani	2				
19	Lee Wei	2				
29	Ravi Shankar	2				
05	Hassan Ibrahim	2				
39	Chong Xin Yi	2				
99	Ravi Bhatt	2				
40	Nadia Zulaikha	2				
43	Mira Anisa	2				
44	Aminah Rafik	2				
52	Zhang Mei	2				
53	Aarav Patel	2				
61	Yang Lei	2				

9.0.12 The total customers that used LaundryBar

Total customer X						
SELECT COUNT(cust_ID) AS ["Total Customer"] FROM CUSTOMER;						
Total customer X						
"Total Cust						
99						

9.0.13 The total customers that used additional minutes specifically for 6 minutes

```
SELECT COUNT(Payment.cust_ID) AS total_customers
FROM (SELECT Payment.cust_ID FROM Payment, Additional_Minutes
WHERE Payment.add_minID = Additional_Minutes.add_minID AND Additional_Minutes.add_minID = 'A1'
GROUP BY Payment.cust_ID) AS SubQuery;
```

9.0.14 The total customers that used Washer 2 from March to December 2024

```
SELECT COUNT(*) AS total_W2_washers
FROM Payment, Machine
WHERE Payment.mach_ID = Machine.mach_ID
AND Payment.mach_ID = 'W2'
AND Payment.pay_date >= #2024-03-01#
AND Payment.pay_date < #2024-12-01#;
```

9.0.15 The total payment made by customers for each month

Total payment for each month X					
SELECT FORMAT(Payment.pay_date, "yyyy-mm") AS payment_month, COUNT(*) AS total_payments FROM Payment WHERE Payment.pay_date IS NOT NULL GROUP BY FORMAT(Payment.pay_date, "yyyy-mm");					
Total payment for each month X					
payment_month		total_payments			
2024-01		8			
2024-02		21			
2024-03		12			
2024-04		5			
2024-05		14			
2024-06		9			
2024-07		7			
2024-08		19			
2024-09		7			
2024-10		3			
2024-11		7			
2024-12		18			

9.0.16 The total payment made by selected customers based on their ID

Total payment for selected cust X					
SELECT COUNT(*) AS total_payments FROM Payment WHERE cust_ID = '06';					
Total payment for selected cust X					
total_payments					
3					

9.0.17 The total sales of LaundryBar

The screenshot shows a SQL query editor with the following text:

```
SELECT SUM(pay_grandTotal) AS total_sales
FROM Payment;
```

Below the query, the results are displayed in a table. The first column is labeled 'total_sales' and contains the value 'RM461.00'.

total_sales
RM461.00

9.0.18 The total sales for each day from the start to the present day

Total sales each day of the week X					
SELECT FORMAT(pay_date, "dddd") AS day_of_week, SUM(pay_grandTotal) AS total_payment FROM Payment GROUP BY FORMAT(pay_date, "dddd");					
Total sales each day of the week X					
day_of_week	total_payment				
Friday	RM54.00				
Monday	RM105.00				
Saturday	RM46.00				
Sunday	RM80.00				
Thursday	RM57.00				
Tuesday	RM78.00				
Wednesday	RM41.00				

9.0.19 The total sales for each machine in LaundryBar

Total sales each machine
<pre> SELECT Machine.mach_ID, SUM(Payment.pay_grandTotal) AS total_sales FROM Machine, Payment WHERE Machine.mach_ID = Payment.mach_ID AND Machine.mach_ID IN ('W1', 'W2', 'W3', 'W4', 'W5', 'D1', 'D2', 'D3', 'D4', 'D5') GROUP BY Machine.mach_ID; </pre>

Total sales each machine					
mach_ID	total_sales				
D1	RM19.00				
D2	RM23.00				
D3	RM31.00				
D4	RM41.00				
D5	RM26.00				
W1	RM74.00				
W2	RM56.00				
W3	RM71.00				
W4	RM47.00				
W5	RM73.00				

9.0.20 The total sales for each machine's temperature in LaundryBar

Total sales each temperature
<pre> SELECT Temperature.temp_type, SUM(Payment.pay_grandTotal) AS total_sales FROM Temperature, Payment WHERE Temperature.temp_ID = Payment.temp_ID GROUP BY Temperature.temp_type; </pre>

Total sales each temperature					
temp_type	total_sales				
High	RM224.00				
Low	RM103.00				
Medium	RM134.00				

9.0.21 Hours with most transaction made by customers

```
Peak hour X
1 SELECT FORMAT(PAYMENT.pay_time, "hh:00 AM/PM") AS hour_slot, COUNT(*) AS total_transactions
2 FROM PAYMENT
3 GROUP BY FORMAT(PAYMENT.pay_time, "hh:00 AM/PM")
4 HAVING COUNT(*) = (
5     SELECT MAX(transaction_count)
6     FROM (
7         SELECT COUNT(*) AS transaction_count
8         FROM PAYMENT
9         GROUP BY FORMAT(PAYMENT.pay_time, "hh:00 AM/PM")
10    )
11 );
```

hour_slot	total_transactions			
05:00 PM	13			

9.0.22 Calculation for pay_grandTotal

```
Query Calculation X
1 SELECT p.pay_ID, ([t].[temp_price] + ([am].[add_minTime] / 6)) AS pay_grandTotal INTO tempPayment
2 FROM Payment AS p, Additional_Minutes AS am, Temperature AS t
3 WHERE p.add_minID = am.add_minID
4 AND p.temp_ID = t.temp_ID;
5
```

9.0.23 Update query calculation

```
Update query calculation X
1 UPDATE PAYMENT
2 INNER JOIN tempPayment ON Payment.pay_ID = tempPayment.pay_ID SET PAYMENT.pay_grandTotal = tempPayment.pay_grandTotal;
3
```

10.0 Appendix

10.1 SQL Query Create Table and Output

CUSTOMER

```
CREATE TABLE CUSTOMER (  
  cust_ID TEXT(10) PRIMARY KEY,  
  cust_name TEXT(50),  
  cust_phoneNum TEXT(15),  
  cust_email TEXT(100),  
  trans_ID TEXT(10),  
  FOREIGN KEY (trans_ID) REFERENCES TRANSACTION(trans_ID)  
);
```

CUSTOMER X						
	cust_ID ▾	cust_name ▾	cust_phoneNum ▾	cust_email ▾	trans_ID ▾	Click to Add ▾
*						

TRANSACTION

```
CREATE TABLE TRANSACTION (  
  trans_ID TEXT(10) PRIMARY KEY,  
  trans_token INTEGER,  
  trans_price CURRENCY  
);
```

TRANSACTION X				
	trans_ID ▾	trans_token ▾	trans_price ▾	Click to Add ▾
*				

MACHINE

```
CREATE TABLE MACHINE (  
  mach_ID TEXT(10) PRIMARY KEY,  
  mach_category TEXT(50),  
  staff_ID TEXT(10),  
  FOREIGN KEY (staff_ID) REFERENCES STAFF(staff_ID)  
);
```

MACHINE X						
	mach_ID ▾	mach_category ▾	staff_ID ▾	Click to Add ▾		
*						

STAFF

```
CREATE TABLE STAFF (  
  staff_ID TEXT(10) PRIMARY KEY,  
  staff_name TEXT(50),  
  staff_email TEXT(100),  
  staff_phoneNum TEXT(15)  
);
```

STAFF X						
	staff_ID ▾	staff_name ▾	staff_email ▾	staff_phone ▾	Click to Add ▾	
*						

PAYMENT

```
CREATE TABLE PAYMENT (
  temp_ID TEXT(10),
  cust_ID TEXT(10),
  mach_ID TEXT(10),
  add_minID TEXT(10),
  FOREIGN KEY (temp_ID) REFERENCES TEMPERATURE(temp_ID),
  FOREIGN KEY (cust_ID) REFERENCES CUSTOMER(cust_ID),
  FOREIGN KEY (mach_ID) REFERENCES MACHINE(mach_ID),
  FOREIGN KEY (add_minID) REFERENCES ADDITIONAL_MINUTES(add_minID)
);
```

	pay_ID ▾	pay_date ▾	pay_time ▾	pay_grandTr ▾	temp_ID ▾	cust_ID ▾	mach_ID ▾	add_minID ▾	Click to Add ▾
*									

TEMPERATURE

```
CREATE TABLE TEMPERATURE (
  temp_ID TEXT(10) PRIMARY KEY,
  temp_type TEXT(10),
  temp_price NUMBER
);
```

	temp_ID ▾	temp_type ▾	temp_price ▾	Click to Add ▾		
*						

ADDITIONAL_MINUTES

```
CREATE TABLE ADDITIONAL_MINUTES (
  add_minID TEXT(10) PRIMARY KEY,
  add_minTime INTEGER
);
```

	add_minID ▾	add_minTim ▾	Click to Add ▾		
*					

10.2 SQL Query Insert Data

CUSTOMER

 **INSERT CUSTOMER** ✕

```
INSERT INTO CUSTOMER
VALUES ('01', 'Ahmad Ali', '010-1234567', 'ahmad.ali@gmail.com', 'TR01');
```

TRANSACTION

 **INSERT TRANSACTION** ✕

```
INSERT INTO TRANSACTION
VALUES ('TR01', 1, 1.00);
```

MACHINE

 **INSERT MACHINE** ✕

```
INSERT INTO MACHINE
VALUES ('D1','Dryer','S01');
```

STAFF

 **INSERT STAFF** ✕

```
INSERT INTO STAFF
VALUES ('S01','Wan Zulkifli','wan1@gmail.com','0126537786');
```

PAYMENT

 INSERT PAYMENT X
INSERT INTO PAYMENT VALUES ('P01',12/8/2024,'3:52:02 AM',5.00,'TH','D1','A1');

TEMPERATURE

 INSERT TEMPERATURE X
INSERT INTO TEMPERATURE VALUES ('TH','High',4);

ADDITIONAL_MINUTES

 INSERT ADDITIONAL_MINUTES X
INSERT INTO ADDITIONAL_MINUTES VALUES ('A1',6);